

A Pell-Group Descent for

$$y^2 = 48x^4 + 12x^2 + 1$$

Theorem 1. *For integers x, y satisfying*

$$y^2 = 48x^4 + 12x^2 + 1,$$

there is no solution with $x \neq 0$. Equivalently, the only integer solutions are $(x, y) = (0, 1), (0, -1)$.

The argument is self-contained: the two elementary facts used throughout are proved in Section 1, and every property of the Pell sequence (u_n, v_n) that is used is proved in Section 2.

1 Two elementary facts

Fact A. *Let a, b be positive integers with $\gcd(a, b) = 1$ such that ab is a perfect square. Then a and b are perfect squares.*

Proof. Write $ab = z^2$. For every prime p the number $\text{ord}_p(a) + \text{ord}_p(b) = 2\text{ord}_p(z)$ is even. Since $\gcd(a, b) = 1$, at least one of $\text{ord}_p(a), \text{ord}_p(b)$ is 0, so the other one is even as well. Thus every prime occurs in a , and in b , with an even exponent; that is, a and b are squares. $\square$

Fact B. (i) *Every square is $\equiv 0$ or $1 \pmod{4}$; in particular no integer $\equiv 2 \pmod{4}$ is a square.*
(ii) *Every square is $\equiv 0, 1, 4$ or $9 \pmod{16}$; in particular no integer $\equiv 7 \pmod{16}$ is a square.*

Proof. (i) $(2k)^2 = 4k^2$ and $(2k+1)^2 = 4k(k+1) + 1$.

(ii) Since $(c+8)^2 = c^2 + 16(c+4)$, the residue of c^2 modulo 16 depends only on $c \pmod{8}$. For $c = 0, 1, \dots, 7$ the squares $0, 1, 4, 9, 16, 25, 36, 49$ are congruent to $0, 1, 4, 9, 0, 9, 4, 1 \pmod{16}$. $\square$

2 The Pell sequence

Put $\varepsilon = 2 + \sqrt{3}$; then $\varepsilon(2 - \sqrt{3}) = 4 - 3 = 1$. Since $\sqrt{3}$ is irrational, $a + b\sqrt{3} = a' + b'\sqrt{3}$ with integers a, b, a', b' forces $a = a'$ and $b = b'$; “comparing coefficients” below refers to this. As $\varepsilon^n \in \mathbb{Z}[\sqrt{3}]$, there are unique integers u_n, v_n ($n \geq 0$) with

$$u_n + v_n\sqrt{3} = \varepsilon^n.$$

Proposition 1. *For all integers $n, a, b, m, s \geq 0$:*

(P1) $u_0 = 1, v_0 = 0$, and $u_{n+1} = 2u_n + 3v_n, v_{n+1} = u_n + 2v_n$. Consequently $u_n \geq 1$ for all n , and $v_n \geq 1$ for $n \geq 1$.

(P2) $u_n^2 - 3v_n^2 = 1$; hence $\gcd(u_n, v_n) = 1$.

(P3) (Addition and doubling formulas) $u_{a+b} = u_a u_b + 3v_a v_b$ and $v_{a+b} = u_a v_b + v_a u_b$. In particular

$$u_{2s} = u_s^2 + 3v_s^2 = 2u_s^2 - 1, \quad v_{2s} = 2u_s v_s.$$

(P4) $u_{n+2} = 4u_{n+1} - u_n$ and $v_{n+2} = 4v_{n+1} - v_n$.

(P5) (Parity) If n is even then u_n is odd and v_n is even; if n is odd then u_n is even and v_n is odd.

(P6) $u_n \equiv 2 \pmod{4}$ for every odd n .

(P7) $u_m v_{m+1} - v_m u_{m+1} = 1$.

Proof. (P1) $\varepsilon^0 = 1$, and $\varepsilon^{n+1} = \varepsilon^n \varepsilon = (u_n + v_n \sqrt{3})(2 + \sqrt{3}) = (2u_n + 3v_n) + (u_n + 2v_n)\sqrt{3}$; compare coefficients. If $u_n \geq 1$ and $v_n \geq 0$, then $u_{n+1} = 2u_n + 3v_n \geq 2$ and $v_{n+1} = u_n + 2v_n \geq 1$; induction gives the inequalities.

(P2) For $n = 0$ the claim reads $1 - 0 = 1$. By (P1),

$$(2u + 3v)^2 - 3(u + 2v)^2 = 4u^2 + 12uv + 9v^2 - 3u^2 - 12uv - 12v^2 = u^2 - 3v^2,$$

so $u_n^2 - 3v_n^2$ does not change when n is replaced by $n + 1$; hence it equals 1 for all n . A common divisor of u_n and v_n divides $u_n^2 - 3v_n^2 = 1$.

(P3) Expand $\varepsilon^{a+b} = \varepsilon^a \varepsilon^b = (u_a + v_a \sqrt{3})(u_b + v_b \sqrt{3})$ and compare coefficients. Taking $a = b = s$ gives $u_{2s} = u_s^2 + 3v_s^2$ and $v_{2s} = 2u_s v_s$. By (P2), $3v_s^2 = u_s^2 - 1$, hence $u_{2s} = 2u_s^2 - 1$.

(P4) $\varepsilon^2 = 7 + 4\sqrt{3} = 4(2 + \sqrt{3}) - 1 = 4\varepsilon - 1$. Multiplying by ε^n gives $\varepsilon^{n+2} = 4\varepsilon^{n+1} - \varepsilon^n$; compare coefficients.

(P5) By (P1), $u_{n+1} \equiv v_n$ and $v_{n+1} \equiv u_n \pmod{2}$, because $2u_n$ and $2v_n$ are even and $3v_n \equiv v_n$. So at each step the parities of (u_n, v_n) are swapped. Starting from $(u_0, v_0) = (1, 0) = (\text{odd}, \text{even})$ this gives (odd, even) for even n and (even, odd) for odd n .

(P6) Induction over odd n . First, $u_1 = 2u_0 + 3v_0 = 2$. If $u_n \equiv 2 \pmod{4}$, then by (P4) $u_{n+2} = 4u_{n+1} - u_n \equiv -u_n \equiv -2 \equiv 2 \pmod{4}$.

(P7) By (P1), $u_m v_{m+1} - v_m u_{m+1} = u_m(u_m + 2v_m) - v_m(2u_m + 3v_m) = u_m^2 - 3v_m^2 = 1$, using (P2). $\square$

3 Four lemmas

Lemma 1 (Pell classification). *If $u, v \geq 0$ are integers with $u^2 - 3v^2 = 1$, then $(u, v) = (u_n, v_n)$ for some integer $n \geq 0$.*

Proof. Strong induction on v . If $v = 0$, then $u^2 = 1$, so $u = 1$ and $(u, v) = (u_0, v_0)$.

Let $v \geq 1$, and assume the statement holds for all solutions with smaller second coordinate. We first record two inequalities.

(i) $u > v$ and $2u > 3v$. Since $u^2 = 3v^2 + 1 > 3v^2$ and $u, v \geq 0$, we have $u > \sqrt{3}v$. Hence $u > v$, and $2u > 2\sqrt{3}v > 3v$, because $2\sqrt{3} > 3$ (as $12 > 9$) and $v > 0$.

(ii) $u \leq 2v$. Since $v \geq 1$, $u^2 = 3v^2 + 1 \leq 4v^2$.

Put

$$u' = 2u - 3v, \quad v' = 2v - u.$$

By (i), $u' > 0$, so $u' \geq 1$. By (ii), $v' \geq 0$. Moreover $v' < v$ is equivalent to $v < u$, which is (i). Also

$$u'^2 - 3v'^2 = (4u^2 - 12uv + 9v^2) - 3(4v^2 - 4uv + u^2) = u^2 - 3v^2 = 1.$$

Thus (u', v') is a solution in non-negative integers with $v' < v$, and by the induction hypothesis $(u', v') = (u_k, v_k)$ for some $k \geq 0$. Finally, expanding shows $u' + v'\sqrt{3} = (u + v\sqrt{3})(2 - \sqrt{3})$; multiplying by $2 + \sqrt{3}$ and using $(2 - \sqrt{3})(2 + \sqrt{3}) = 1$ gives

$$u + v\sqrt{3} = (u_k + v_k\sqrt{3})(2 + \sqrt{3}) = \varepsilon^{k+1},$$

i.e. $(u, v) = (u_{k+1}, v_{k+1})$. □

Lemma 2 (Congruence filter). *For $n \geq 0$,*

$$v_n \equiv 0, 1, 4, 7 \pmod{8} \quad \text{according as} \quad n \equiv 0, 1, 2, 3 \pmod{4}.$$

In particular $v_n \equiv 1 \pmod{8}$ if and only if $n \equiv 1 \pmod{4}$.

Proof. Let $w_n = (v_n, v_{n+1}) \pmod{8}$; then $w_0 = (0, 1)$. By (P4), w_{n+1} is obtained from w_n by the map $(a, b) \mapsto (b, 4b - a)$, and

$$(0, 1) \mapsto (1, 4) \mapsto (4, 15 \equiv 7) \mapsto (7, 24 \equiv 0) \mapsto (0, -7 \equiv 1) = (0, 1).$$

So $w_4 = w_0$, hence $w_{n+4} = w_n$ for all n , and the first coordinates run through $0, 1, 4, 7$ periodically. □

Lemma 3 (Halving identity). *For every $m \geq 0$,*

$$v_{2m+1} - 1 = 2v_m u_{m+1}.$$

Proof. By (P3) with $(a, b) = (m, m+1)$, $v_{2m+1} = u_m v_{m+1} + v_m u_{m+1}$. By (P7), $u_m v_{m+1} = 1 + v_m u_{m+1}$. Hence $v_{2m+1} = 1 + 2v_m u_{m+1}$. □

Lemma 4 (Key descent). *For every integer $r \geq 1$, the numbers u_r and v_r are not both perfect squares.*

Proof. Suppose the contrary, and let $r \geq 1$ be the *smallest* index for which $u_r = c^2$ and $v_r = d^2$ with integers c, d . Every $r \geq 1$ is odd, or $\equiv 2 \pmod{4}$, or $\equiv 0 \pmod{4}$.

Case 1: r odd. By (P6), $u_r \equiv 2 \pmod{4}$, which is not a square by Fact B(i). Contradiction.

Case 2: $r \equiv 2 \pmod{4}$. Write $r = 2s$ with s odd. By (P3), $u_r = 2u_s^2 - 1$. By (P6), $u_s = 4k + 2$ for some integer k , so $u_s^2 = 16k^2 + 16k + 4 \equiv 4 \pmod{16}$ and therefore $u_r \equiv 2 \cdot 4 - 1 = 7 \pmod{16}$. By Fact B(ii), u_r is not a square. Contradiction.

Case 3: $r \equiv 0 \pmod{4}$. Write $r = 4s$ with $s \geq 1$, and put $t = 2s$. By (P3),

$$d^2 = v_r = v_{2t} = 2u_tv_t = u_t \cdot (2v_t).$$

Since t is even, u_t is odd by (P5); together with $\gcd(u_t, v_t) = 1$ (P2) this gives $\gcd(u_t, 2v_t) = 1$. Both factors are positive (P1), so by Fact A each is a square; in particular $2v_t = e^2$. Then e is even, say $e = 2f$, and $v_t = 2f^2$. On the other hand (P3) gives $v_t = v_{2s} = 2u_sv_s$, hence

$$u_sv_s = f^2.$$

Since $\gcd(u_s, v_s) = 1$ and $u_s, v_s \geq 1$, Fact A shows that u_s and v_s are both squares. But $1 \leq s = r/4 < r$, contradicting the minimality of r .

All cases lead to a contradiction, which proves the lemma. $\square$

Remark. By Lemma 1, the pairs (c^2, d^2) with $c^4 - 3d^4 = 1$ and $d \neq 0$ are exactly the pairs (u_r, v_r) , $r \geq 1$, in which both entries are squares. So Lemma 4 is equivalent to the statement that $c^4 - 3d^4 = 1$ has no integer solution with $d \neq 0$.

4 Proof of Theorem 1

If $x = 0$, then $y^2 = 1$, so $(x, y) = (0, \pm 1)$, and these are solutions. It remains to rule out solutions with $x \neq 0$. Suppose (x, y) is one. Only x^2 and y^2 occur in the equation, so we may replace (x, y) by $(|x|, |y|)$ and assume $x \geq 1$ and $y \geq 0$.

Step 1 (Pell envelope). We have

$$4y^2 - 3(8x^2 + 1)^2 = (192x^4 + 48x^2 + 4) - (192x^4 + 48x^2 + 3) = 1.$$

Hence, with $u = 2y$ and $v = 8x^2 + 1$,

$$u^2 - 3v^2 = 1, \quad u, v \geq 0, \quad v \equiv 1 \pmod{8}, \quad v \geq 9.$$

Step 2 (index of v). By Lemma 1, $(u, v) = (u_n, v_n)$ for some $n \geq 0$. By Lemma 2, $n \equiv 1 \pmod{4}$. Moreover $n \neq 1$, because $v_1 = 1 < 9 \leq v$. So $n \geq 5$. Write $n = 2m + 1$; since $n \equiv 1 \pmod{4}$, m is even, and $m \geq 2$.

Step 3 (halving). By Lemma 3, $8x^2 = v_{2m+1} - 1 = 2v_mu_{m+1}$, that is,

$$4x^2 = v_mu_{m+1}. \tag{1}$$

Step 4 (a gcd). As m is even, v_m is even; as $m + 1$ is odd, u_{m+1} is even (P5). Let $g = \gcd(v_m, u_{m+1})$. By (P1), $u_{m+1} - 3v_m = 2u_m$, so $g \mid 2u_m$. Also $\gcd(g, u_m)$ divides $\gcd(v_m, u_m) = 1$ (P2). Hence $g \mid 2$, and since both numbers are even, $g = 2$. Dividing (1) by 4,

$$\frac{v_m}{2} \cdot \frac{u_{m+1}}{2} = x^2, \quad \gcd\left(\frac{v_m}{2}, \frac{u_{m+1}}{2}\right) = 1,$$

and both factors are positive integers. By Fact A, $v_m/2 = b^2$ for some integer b .

Step 5 (second halving and contradiction). Write $m = 2r$ with $r \geq 1$. By (P3), $v_m = v_{2r} = 2u_r v_r$, so

$$u_r v_r = b^2.$$

Since $\gcd(u_r, v_r) = 1$ (P2) and $u_r, v_r \geq 1$ (P1), Fact A shows that u_r and v_r are both perfect squares. This contradicts Lemma 4.

Hence no solution with $x \neq 0$ exists, and the only integer solutions are $(x, y) = (0, \pm 1)$. $\square$